

AI Studio Challenge Project Examples

The following examples represent real AI Studio Challenge Projects from our most recent program cycle, organized by ML problem type. They are meant to inspire and guide you as you develop your own project proposal. Each example illustrates how a partner organization framed a real-world business or social challenge as a machine learning problem suitable for undergraduate students.

Large Language Models (LLMs)

Retail & eCommerce

Agentic AI for Insights

Build an intelligent assistant using RAG techniques, LLMs, vector search, and agent orchestration to extract and summarize actionable insights from unstructured enterprise documents.

Technology

CRM Intelligence Assistant

Apply machine learning techniques to synthetic CRM datasets to build a model to predict and analyze sales outcomes and customer behavior patterns – resulting in an interactive chatbot that can assist sales teams.

Business Intelligence

Technology

AI-Powered Trendspotting

Build an AI system that analyzes Google Trends data to automatically detect emerging topics, products, or services; classify their potential growth patterns; and provide insights into their early trajectory.

Food | Beverage | CPG

Flavor Profiling to Identify Emerging Flavor Trends

Use publicly available data to build an ML system to identify trending flavors in food categories; classify consumer sentiment around these flavors; and predict potential success of flavor combinations.

Predictive Modeling

Business Services/Consulting

Predicting Power Outages Using Weather and Cyclone Data

Use National Weather Service API and National Hurricane Center data to build a predictive model for electrical power outages, to understand trends and identify areas needing intervention.

Business Services/Consulting

Predictive Maintenance for Oil & Gas Pump Operation

Use synthetic machine operating and health monitoring data to build a model to predict impending pump failures, to enable preventative maintenance and reduce unplanned downtime.

Recommendation Systems

Technology

AI Assisted Recruitment

Use NLP to automatically match candidates with job positions by analyzing resumes against job requirements, generating match summaries for hiring managers, and providing feedback to rejected applicants.

Business Services/Consulting

Investor & Developer Matching Engine

Use cosine-similarity and recommendation algorithms on EB-5 regional-center directories and startup profiles, to match investors with projects that fit their risk and sector preferences.

Computer Vision

Technology

Instance Segmentation for Autonomous Driving

Build an ML model that identifies and segments individual instances of objects like cars, people, and bikes in driving scenes using datasets such as BDD100K and CityScapes.

Telecommunications

Protecting Avian Wildlife on Cellular Networks

Using ML techniques such as object detection and image classification, build a model that can identify birds and bird nests on cell towers, for wildlife protection and regulatory compliance.

Natural Language Processing

Business Services/Consulting

Fake News Categorization

Use NLP and deep learning to analyze news articles and classify them as real or fake, detect problematic content types, and assign trustworthiness scores.

Technology

Finding Dataset Bias Using Language Models

Use NLP techniques such as word embeddings to analyze social bias in text datasets, to better understand the impact of biased datasets on NLP models.